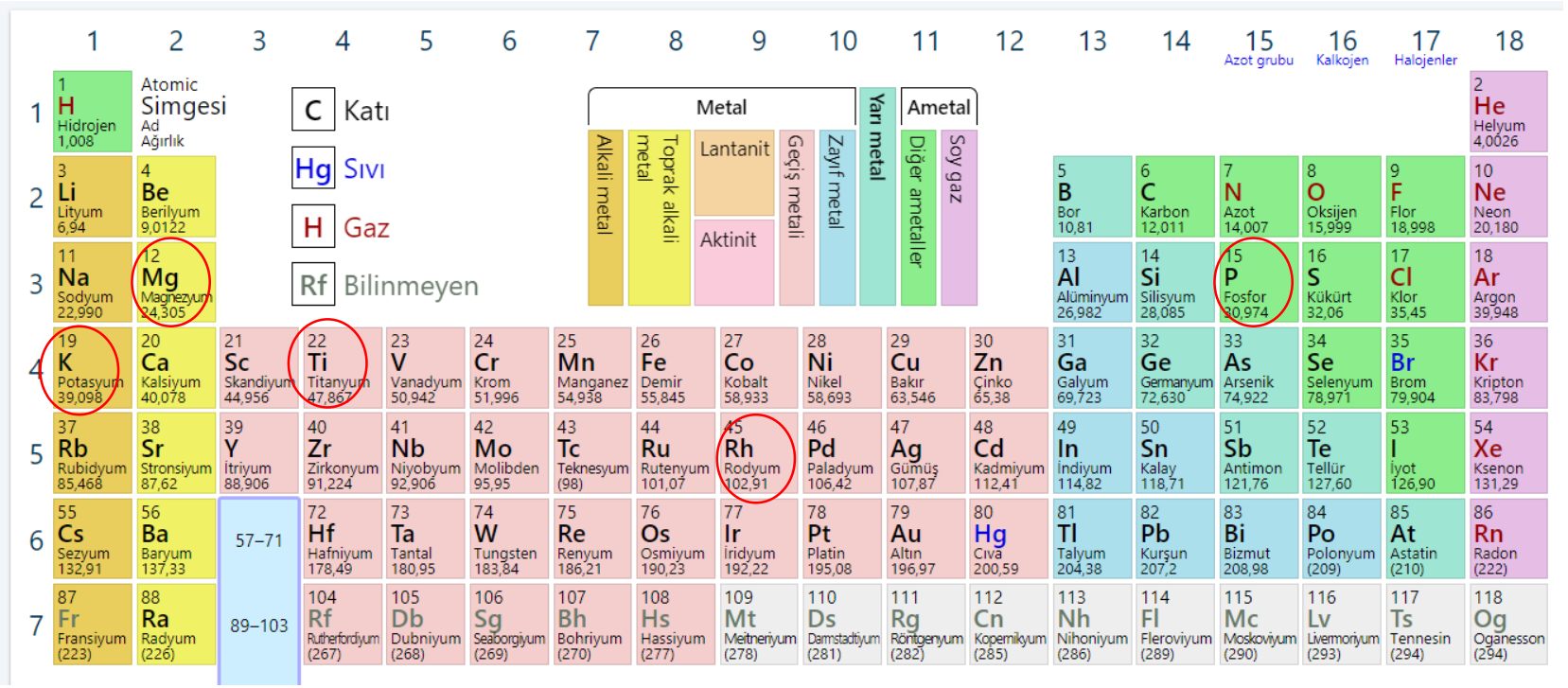
EXERCISES ABOUT CHAPTER 7

Periodic Properties of the Elements

- Arrange the following atoms in order of increasing effective nuclear charge experienced by the electrons in the n=3 electron shell: K, Mg, P, Rh, and Ti.
  Rank from smallest to largest.



The effective nuclear charge increases from left to right across any period of the periodic table.

Ranking Z_{eff} for the electrons in the n=3 electron shell for the elements means that the trend will not necessarily follow the same trend in Z_{eff} observed for the valence shell.

When moving from left to right across a row of the periodic table, the effective nuclear charge increases as the number of protons in the nucleus increases, while the core electrons (most effective at shielding) remain constant. When moving down a column it also increases.

An alternative method for correctly sorting the items in this specific series is to place them in order of increasing atomic number, which in this case correlates to the trends of effective nuclear charge since the Z_{eff} to be ranked is for the electrons in the n=3 electron shell.

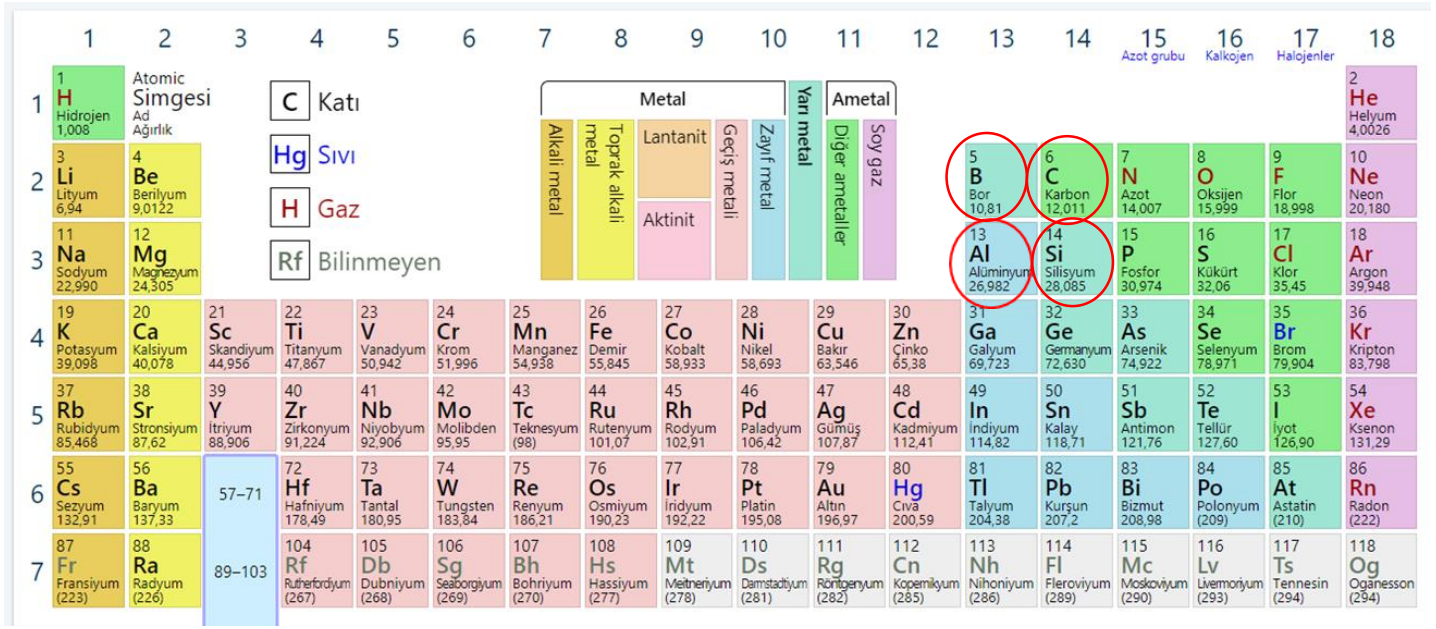
How can we account for the combination of nuclear attraction and electron repulsions for our electron of interest? The simplest way to do so is to imagine that the electron experiences a net attraction that is the result of the nuclear attraction decreased by the electron–electron repulsions. We call this partially screened nuclear charge the **effective nuclear charge**, Z_{eff}. Because the full attractive force of the nucleus has been decreased by the electron repulsions, we see that the effective nuclear charge is always less than the *actual* nuclear charge ($Z_{eff} < Z$). We can define the amount of screening of the nuclear charge quantitatively by using a *screening constant*, S, such that

$$Z_{eff} = Z - S$$

where S is a positive number. For a valence electron, most of the shielding is due to the core electrons, which are much closer to the nucleus. As a result, for the valence electrons in an atom *the value of S is usually close to the number of core electrons in the atom*. (Electrons

Mg < P < K < Ti < Rh

- Referring to the periodic table, arrange (as much as possible) the atoms B, C, Al, and Si in order of increasing size.



increasing radius

increasing radius

radius C < radius B

radius Si < radius Al

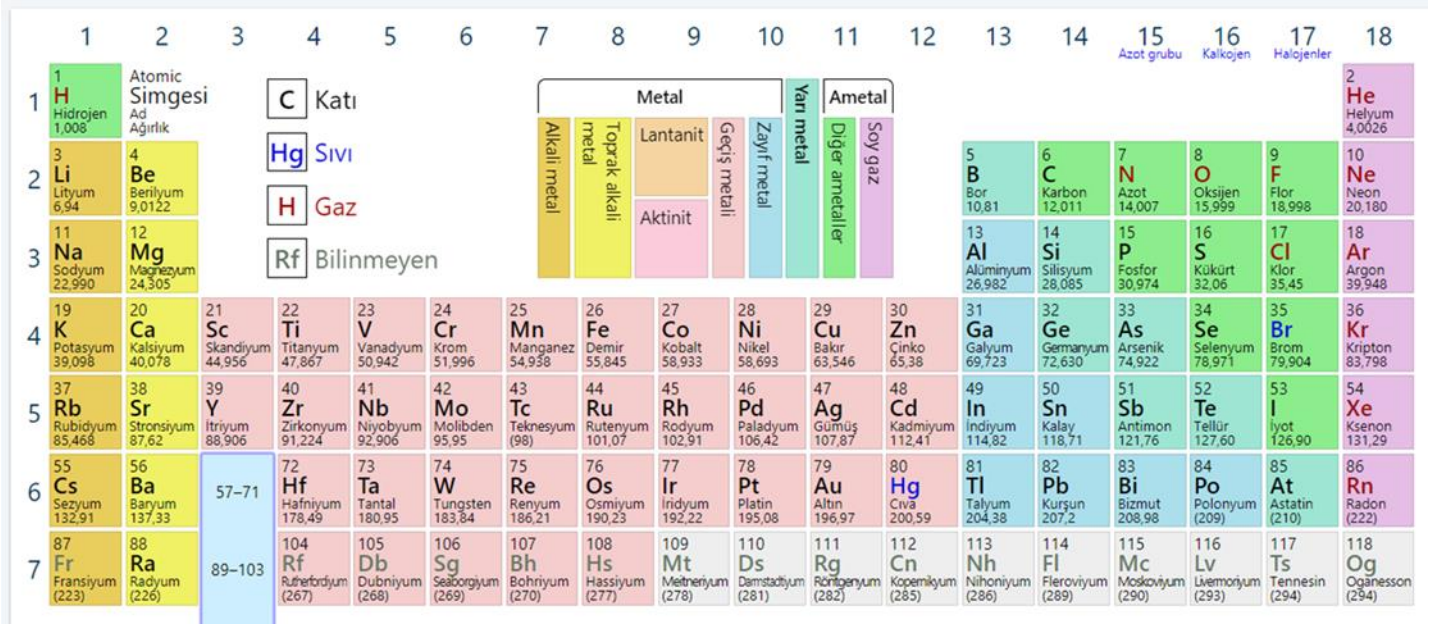
radius B < radius Al
radius C < radius Si

radius C < radius B ~ radius Si < radius Al

- Which of the following is an isoelectronic series?

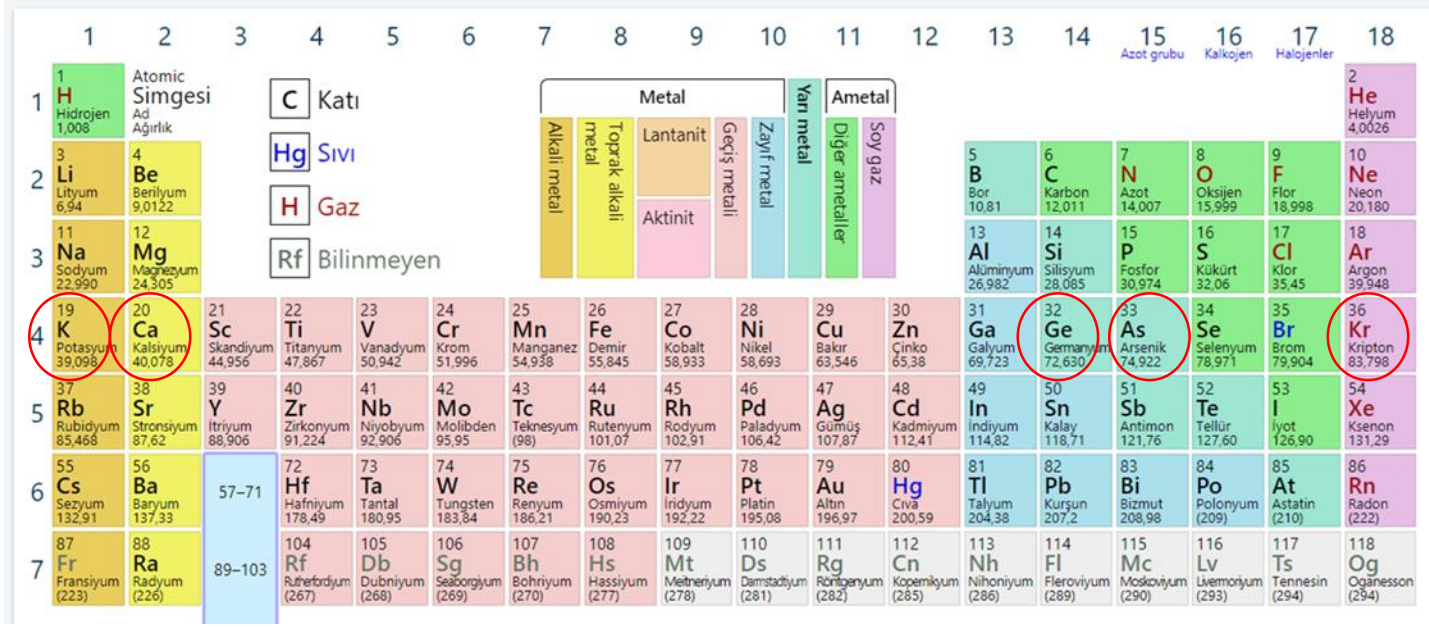
  - F^-, Cl^-, Br^-, I^-
  - S, Cl, Ar, K
  - O^{2-}, F^-, Ne, Na^+
  - Si^{2-}, P^{2-}, S^{2-}, Cl^{2-}
  - B^{5-}, Si^{4-}, As^{3-}, Te^{2-}

O^{2-}, F^{-}, Ne, Na^{+}


Atomic
Simgesi
Ad
Ağırlık
C Katı
Hg Sıvı
H Gaz
Rf Bilinmeyen
Metal
Alkali metal
Toprak alkali metal
Lantanit
Aktinit
Geçiş metali
Zayıf metal
Yarı metal
Ametal
Diğer ametaller
Soy gaz
Azot grubu
Kalkojen
Halojenler
57–71
89–103

- Of the following, which gives the correct order for atomic radius for Ca, K, As, Ge and Kr?

  - ○ Ca > K > As > Ge > Kr
  - ○ K > Ca > Ge > As > Kr
  - ○ Ge > As > Kr > K > Ca
  - ○ Kr > As > Ge > Ca > K
  - ○ Kr > Ge > As > K > Ca



increasing radius

increasing radius

- Consider the following electron configurations to answer the question:

(i) $1s^2\ 2s^2\ 2p^6\ 3s^1$
(ii) $1s^2\ 2s^2\ 2p^6\ 3s^2$
(iii) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$
(iv) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$
(v) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$

The electron configuration that belongs to the atom with the lowest second ionization energy is _____.

- ○ (i)
- ○ (ii)
- ○ (iii)
- ○ (iv)
- ○ (v)

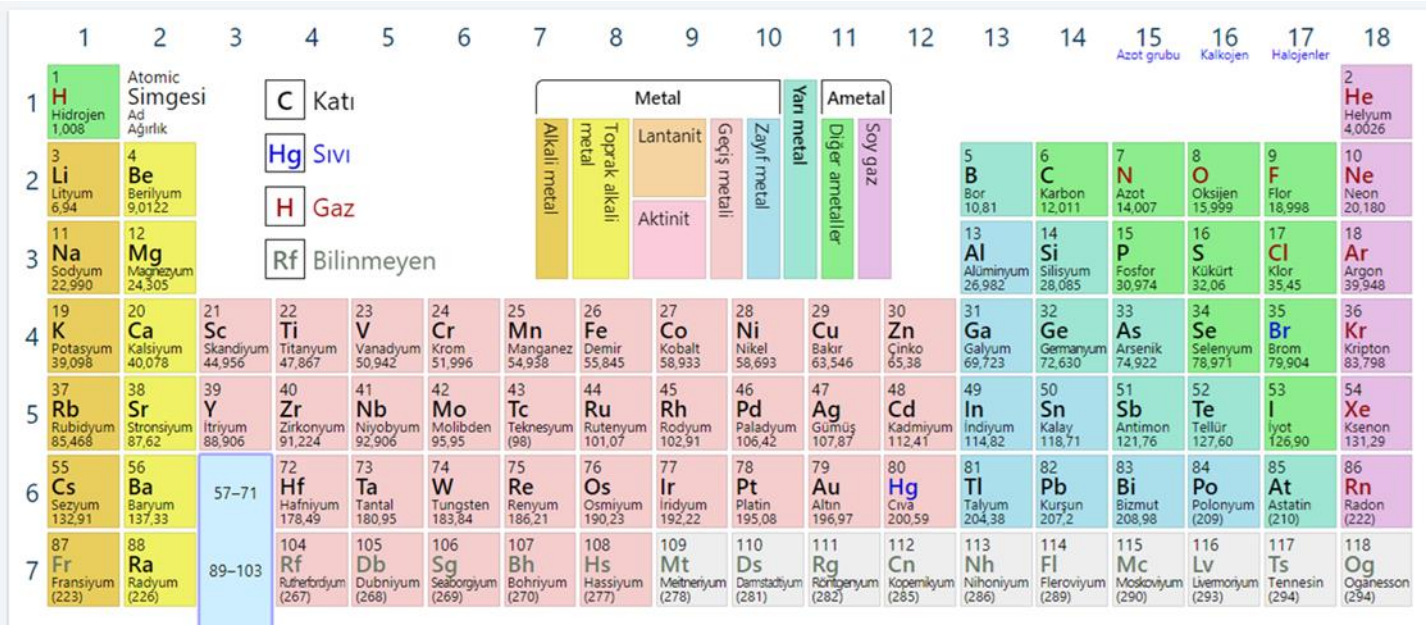
(i) $1s^2\ 2s^2\ 2p^6\ 3s^1$ Na

(ii) $1s^2\ 2s^2\ 2p^6\ 3s^2$ Mg

(iii) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$ Al

(iv) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$ S

(v) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$ Cl



increasing ionization energy

increasing ionization energy

- Consider the following equation:

$$Ca^{+}_{(g)} + e^{-} \rightarrow Ca_{(g)}$$

Which of the following statements are true?

- [ ] The energy change for this process is the negative of the first ionization energy of the Ca atom.
- [ ] The energy change for this process is the electron affinity of the Ca^{+} ion.
- [ ] The energy change for this process is the negative of the electron affinity of the Ca atom.